SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	Brasilla R	**Reg. No.:**	192511357
Date:	14/10/2026	**Duration:**	60 minutes

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. **Identify the Project Vision and Stakeholders**
   - Define the project's vision, objectives, and identify all key stakeholders involved.
2. **Create Epics and User Stories**
   - Break down the requirements into Epics and formulate well-structured User Stories using the format: *As a <user>, I want <feature>, so that <benefit>*.
3. **Define Acceptance Criteria**
   - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
4. **Prioritize the Product Backlog**
   - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
5. **Plan the Sprint Backlog**
   - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
6. **Estimate Effort Using Story Points**
   - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
7. **Present the Sprint Goal and Expected Deliverables**
   - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Problem Statement:

Vehicle Service and Maintenance Management System: Develop a system that allows vehicle owners to book service appointments, maintain service history, receive maintenance reminders, and track service status.

Agile Sprint Planning and User Story Workshop

Project Title: Vehicle Service and Maintenance Management System

1. Project Vision and Stakeholders

The vision of the Vehicle Service and Maintenance Management System is to develop a user-friendly application that helps vehicle owners manage their vehicle servicing activities easily. The system allows users to book service appointments, maintain service history, receive maintenance reminders, and track the current status of their vehicles. The main objective is to reduce manual work, avoid missing service dates, and improve customer satisfaction. The key stakeholders involved in this project are vehicle owners, service centre staff, service managers, system administrators, and software developers. Vehicle owners use the system to manage their services, while service centre staff update service details and managers monitor the overall service activities.

2. Create Epics and User Stories

Epics represent the major features of the system, while user stories describe the requirements from the user's point of view.

Epic 1: Vehicle and User Management. As a vehicle owner, I want to register and add my vehicle details, so that I can manage my vehicle information in one place. As a vehicle owner, I want to log in securely, so that my personal and vehicle information remains protected.

Epic 2: Service Appointment Booking. As a vehicle owner, I want to book a service appointment by selecting a date, time, and service type, so that I can get my vehicle serviced conveniently. As a service centre staff member, I want to view and manage appointment bookings, so that I can organize service schedules efficiently.

Epic 3: Service History Management. As a vehicle owner, I want to view my previous service records, so that I can understand the maintenance history of my vehicle. As a service centre staff member, I want to update completed service details, so that accurate maintenance records are available.

Epic 4: Maintenance Reminders. As a vehicle owner, I want to receive reminders for upcoming services, so that I do not miss important maintenance dates.

Epic 5: Service Status Tracking. As a vehicle owner, I want to track the current status of my vehicle service, so that I know whether my vehicle is waiting, under service, or ready for collection.

3. Define Acceptance Criteria

Acceptance criteria are the conditions that must be satisfied for a user story to be considered complete.

For vehicle registration, Given that a vehicle owner has logged in, When they enter valid vehicle details, Then the system must save the details and display the registered vehicle.

For appointment booking, Given that a vehicle owner has selected a service type and an available time slot, When they confirm the booking, Then the system must create the appointment and display its booking details.

For service history, Given that a vehicle has completed service records, When the owner opens the service history page, Then the system must display the previous service dates, service types, and costs.

For maintenance reminders, Given that a vehicle has an upcoming service due date, When the reminder date is reached, Then the system must notify the vehicle owner about the upcoming maintenance.

For service status tracking, Given that a service appointment has been created, When the service centre staff updates its status, Then the latest status must be displayed to the vehicle owner.

4. Prioritize the Product Backlog

The product backlog contains all the features and requirements arranged according to their importance, business value, and dependencies. The MoSCoW prioritization technique is used to classify requirements as Must Have, Should Have, Could Have, and Won't Have for the current release. User registration, vehicle details management, and service appointment booking are classified as Must Have because they form the basic functionality of the system. Service status tracking and service history are classified as Should Have because they improve service transparency and record management. Advanced analytics, online payment integration, and additional notification options can be classified as Could Have. Features that are not essential for the initial release may be postponed. The backlog is reviewed regularly and updated based on stakeholder feedback.

5. Plan the Sprint Backlog

The first sprint focuses on developing the basic features required to start using the system. The selected user stories include user registration, vehicle details management, and service appointment booking. The development tasks include designing the registration and login pages, creating the vehicle details form, designing the appointment booking interface, developing backend APIs, creating the database tables, validating user inputs, and testing the booking process. The team also performs integration testing to ensure that vehicle details and appointment information are saved correctly. These tasks are assigned to team members based on their responsibilities and completed within the sprint duration.

User Story	**Story Points**	**Justification**
User registration and login	5	Requires form validation, authentication, and basic security.
Add and manage vehicle details	3	Involves a simple form and database operations.

Book service appointments	8	Requires date and time selection, availability checks, and booking storage.
View service history	5	Requires retrieving and displaying previous service records.
Track service status	5	Requires status updates and displaying the latest information.
Receive maintenance reminders	8	Requires due-date checking and notification functionality.

6. Estimate Effort Using Story Points

The team uses the Fibonacci sequence (1, 2, 3, 5, 8, 13) to estimate the effort required for each user story through Planning Poker. Story points represent the relative complexity, effort, and uncertainty of a task rather than the exact number of working hours.

These estimates are preliminary and may be adjusted after team discussion. The first sprint can include the first three user stories, with a total estimated effort of 16 story points. The final selection depends on the team's capacity and the complexity discovered during development.

7. Sprint Goal and Expected Deliverables

The goal of the first sprint is to develop the basic vehicle service management functionality, allowing users to register, add their vehicle details, and book service appointments successfully. At the end of the sprint, the team is expected to deliver a working registration and login module, a vehicle details management module, and a service appointment booking module connected to a database. The system should validate user inputs, save vehicle and booking information, and display appointment confirmation. The team will demonstrate the completed features during the Sprint Review, collect feedback from stakeholders, and identify improvements during the Sprint Retrospective. The remaining features, including service history, maintenance reminders, and service status tracking, will be planned for subsequent sprints.

Conclusion

Agile Sprint Planning helps the development team divide the Vehicle Service and Maintenance Management System into small, manageable tasks. By identifying stakeholders, creating epics and user stories, defining acceptance criteria, prioritizing the backlog, and estimating effort using story points, the team can develop the system in an organized and iterative manner. Regular reviews and feedback help ensure that the final system meets the needs of vehicle owners and service centre staff.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification **(30%)**	/5
Epics & User Stories **(25%)**	/5
Acceptance Criteria **(20%)**	/5
Product Backlog Prioritization & Sprint Planning **(15%)**	/5
Story Point Estimation, Sprint Goal & Presentation **(10%)**	/5
TOTAL	/25